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CHAPTER 5

INTERPOLATION WITH EQUALLY SPACED DATA

5.1 THE DIFFERENCE TABLE

Usually the data is available at equal spacing of argument. In this case it is useful to form a difference table. To get an entry $\Delta^r y_k$, say, in the table we only have to subtract $\Delta^{r-1} y_k$ from $\Delta^{r-1} y_{k+1}$ i.e. we find the difference of the two entries immediately to the left.

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
x_0	y_0	Δy_0					
x_1	y_1		$\Delta^2 y_0$				
		Δy_1		$\Delta^3 y_0$			
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$		
		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$	
x_3	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$		$\Delta^6 y_0$
		Δy_3		$\Delta^3 y_2$		$\Delta^5 y_1$	
x_4	y_4		$\Delta^2 y_3$		$\Delta^4 y_2$		
		Δy_4		$\Delta^3 y_3$			
x_5	y_5		$\Delta^2 y_4$				
		Δy_5					
x_6	y_6						

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INTERPOLATION WITH EQUALLY SPACED DATA

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x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
x_0	y_0	Δy_0					
x_1	y_1	Δy_1	$\Delta^2 y_0$	$\Delta^3 y_0$			
x_2	y_2	Δy_2	$\Delta^2 y_1$	$\Delta^3 y_1$	$\Delta^4 y_0$	$\Delta^5 y_0$	
x_3	y_3	Δy_3	$\Delta^2 y_2$	$\Delta^3 y_2$	$\Delta^4 y_1$	$\Delta^5 y_1$	$\Delta^6 y_0$
x_4	y_4	Δy_4	$\Delta^2 y_3$	$\Delta^3 y_3$	$\Delta^4 y_2$		
x_5	y_5	Δy_5	$\Delta^2 y_4$				
x_6	y_6						

We see that the quantities $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0, \dots, \Delta^6 y_0$ lie on a diagonal starting at y_0 and sloping downwards to the right.

Similarly $\Delta y_2, \Delta^2 y_2, \Delta^3 y_2, \dots, \Delta^4 y_2$ lie on the diagonal starting at y_2 and sloping downwards to the right. Now if we consider a diagonal starting at y_4 and sloping upwards to the right, we see that $\Delta y_3, \Delta^2 y_2, \Delta^3 y_1, \Delta^4 y_0$ lie on this diagonal.

But since $\Delta = E\nabla$, we have

$$\begin{aligned}\Delta y_3 &= E\nabla y_3 = \nabla y_4, \\ \Delta^2 y_2 &= (E\nabla)^2 y_2 = \nabla^2 y_4, \\ \Delta^3 y_1 &= (E\nabla)^3 y_1 = \nabla^3 y_4, \\ \Delta^4 y_0 &= (E\nabla)^4 y_0 = \nabla^4 y_4.\end{aligned}$$

Thus we see that the elements lying on a diagonal starting at y_4 and sloping upwards to the right are the backward differences $\nabla y_4, \nabla^2 y_4, \nabla^3 y_4, \dots$, and the same is true for every entry in the table. Now the elements on the horizontal line starting at y_3 are $\Delta^2 y_2, \Delta^4 y_1$, and $\Delta^6 y_0$. But since $\Delta^2 = E\delta^2$, we have

$$\begin{aligned}\Delta^2 y_2 &= E\delta^2 y_2 = \delta^2 y_3, \\ \Delta^4 y_1 &= (E\delta^2)^2 y_1 = \delta^4 y_3, \\ \Delta^6 y_0 &= (E\delta^2)^3 y_0 = \delta^6 y_3.\end{aligned}$$

Thus we see that the quantities $\delta^2 y_3, \delta^4 y_3, \dots$ lie on a horizontal straight line starting at y_3 and the same is true about any other entry.

EXAMPLE 5.1: Form a difference table from the data given below and read $\Delta^2(2.3)$, $\Delta^2(4.2)$, $\Delta^2(4.2)$ and $\Delta^4(4.2)$.

	x	y
x_0	1.0	2.0
x_1	1.2	3.5
x_2	1.4	-1.7
x_3	1.6	2.3
x_4	1.8	4.2
x_5	2.0	6.5
x_6	2.2	5.7

SOLUTION: The difference table will be as follows:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
1.0	2.0						
		1.5					
1.2	3.5		-6.7				
		-5.2		15.9			
1.4	-1.7		9.2		-27.2		
		4.0		-11.3		41.0	
1.6	2.3		-2.1		13.8		-60.8
		1.9		2.5		-19.8	
1.8	4.2		0.4		-6.0		
		2.3		-3.5			
2.0	6.5		-3.1				
		-0.8					
2.2	5.7						

To find $\Delta^2(2.3)$ we look along the diagonal starting at 2.3 and sloping downwards. On this diagonal the successive entries are $\Delta(2.3)$, $\Delta^2(2.3)$, $\Delta^3(2.3)$ and so on. Thus

$$\Delta^2(2.3) = 0.4.$$

On the other hand the entries on the diagonal starting at 4.2 and

sloping upwards are $\nabla(4.2)$, $\nabla^2(4.2)$ and so on. Thus

$$\nabla^2(4.2) = -2.1.$$

Finally, to find $\delta^2(4.2)$ and $\delta^4(4.2)$ we look along the horizontal line starting at 4.2. Thus

$$\delta^2(4.2) = 0.4,$$

$$\delta^4(4.2) = -6.0.$$

Now consider Δx^k where k is a positive integer.

$$\begin{aligned}\Delta x^k &= (x+h)^k - x^k \\ &= (x^k + kx^{k-1}h + \dots + h^k) - x^k \\ &= P_{k-1}(x)\end{aligned}$$

where $p_{k-1}(x)$ is a polynomial of degree $k-1$.

Let $Q_n(x)$ be a polynomial of degree n :

$$Q_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0, \quad a_n \neq 0.$$

Now

$$\begin{aligned}\Delta Q_n(x) &= \Delta(a_n x^n + a_{n-1} x^{n-1} + \dots + a_0), \\ &= a_n \Delta x^n + a_{n-1} \Delta x^{n-1} + \dots + \Delta a_0,\end{aligned}$$

since Δ is a linear operator. Thus we have

$$\begin{aligned}\Delta Q_n(x) &= a_n (\text{a polynomial of degree } n-1) + \\ &\quad a_{n-1} (\text{a polynomial of degree } n-2) + \dots + 0.\end{aligned}$$

It is clear that $\Delta Q_n(x)$ will be a polynomial of degree $n-1$. If we operate twice with Δ on $Q_n(x)$ the result will be a polynomial of degree $n-2$. If we continue like this we see that $\Delta^n Q_n(x)$ will be a constant namely $n!a_n$ independent of x . If we are dealing with data pertaining to $y = Q_n(x)$, all entries appearing in the column under the heading Δ^n will be same and consequently all the succeeding columns will consist entirely of zeros. Thus we have proved:

THEOREM 5.1: The n -th differences of an n -th degree polynomial are constant and all the higher differences are zero.

5.2 NEWTON'S FORWARD AND BACKWARD DIFFERENCE FORMULAE

Suppose we are given the values $\dots y_{-2}, y_{-1}, y_0, y_1, y_2, \dots$ of a function for $x = \dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots$, where $x_{r+1} - x_r = h (r = \dots, -2, -1, 0, 1, \dots)$ and we wish to find $f(x_0 + ph)$ where in general $-1 < p < 1$. Now we have found that

$$P_n(x) = f(x_0) + (x-x_0)f(x_0, x_1) + (x-x_0)(x-x_1)f(x_0, x_1, x_2) + \dots + (x-x_0)(x-x_1)\dots(x-x_{n-1})f(x_0, x_1, \dots, x_n)$$

is the polynomial that interpolates the given function at the points (x_r, y_r) , $r = 0, 1, \dots, n$. Now

$$f(x_0, x_1) = \frac{\Delta y_0}{h},$$

$$f(x_0, x_1, x_2) = \frac{\Delta^2 y_0}{2!h^2}$$

$$\dots$$

$$f(x_0, x_1, \dots, x_n) = \frac{\Delta^n y_0}{n!h^n}$$

Also if we let $x = x_0 + ph$ then

$$x - x_0 = ph$$

$$x - x_1 = x - x_0 - (x_1 - x_0) = ph - h = h(p - 1)$$

$$x - x_2 = x - x_1 - (x_2 - x_1) = h(p - 1) - h = h(p - 2)$$

$$\dots$$

$$x - x_n = h(p - n).$$

Substitution in $P_n(x)$ gives

$$y_p \approx P_n(x_0 + ph) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \dots + \frac{p(p-1)\dots(p-n+1)}{n!} \Delta^n y_0 \quad (5.1)$$

This is the Newton's forward difference formula for equally spaced data.

ALTERNATIVE DERIVATION:

Using the operator notation we have

$$\begin{aligned} y_p &= E^p y_0 = (1 + \Delta)^p y_0 \\ &= y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots \end{aligned}$$

which is the same formula as above.

NEWTON'S BACKWARD DIFFERENCE FORMULA:

We have

$$\begin{aligned} y_p &\approx (E^{-1})^{-p} y_0 = (1 - \nabla)^{-p} y_0 \\ &= y_0 + p\nabla y_0 + \frac{p(p+1)}{2!} \nabla^2 y_0 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_0 + \dots \quad (5.2) \end{aligned}$$

This is called Newton's backward difference formula. The two Newton's formulae are usually used for interpolation near the beginning or the end of a table. If we are interested in a value in the middle, we should make use of a formula which uses values symmetrically placed about y_0 . We shall discuss such formulae in the next section.

EXAMPLE 5.2: The following table lists values of the function $y = \sin(x^2 + 1) + \cos x$ at certain points. Estimate $y(-0.4)$ from the given data.

x	-0.5	-0.3	-0.1	0.1	0.3	0.5
y	1.8265671	1.8419634	1.841836	1.841836	1.8419634	1.8265671

SOLUTION: Since the point -0.4 lies near the beginning of the table, Newton's forward difference formula would be appropriate. The difference table is as follows:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
-0.5	1.8265671					
		0.0153963				
-0.3	1.8419634		-0.0155237			
		-0.0001274		0.0156511		
-0.1	1.8418360		0.0001274		-0.0156511	
		0.0000000		0.0000000		0
0.1	1.8418360		0.0001274		-0.0156511	
		0.0001274		-0.0156511		
0.3	1.8419634		-0.0155237			
		-0.0153963				
0.5	1.8265671					

Now $f(x_0 + ph) = y_p \approx y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots$

Here $x_0 = -0.5$, $h = 0.2$ and $x = -0.4$.

But we have $x = x_0 + ph$, so

$$-0.4 = -0.5 + p(0.2)$$

or $p = 0.5$.

Let $p = 0.5$ and substitute for y_0 , Δy_0 , $\Delta^2 y_0$ etc. We have

$$f(-0.4) \approx 1.8378641.$$

Compare this with the exact value of $\sin(x^2+1) + \cos x$ at -0.4 which is 1.8378....

5.3 THE LOZENGE DIAGRAM

In this section we introduce the lozenge diagram which is very useful in finding interpolation formulae of various kinds. For this purpose we introduce the notation

$$C(p, n) = \frac{p(p-1)(p-2)\dots(p-n+1)}{n!}$$

In this notation Newton's forward difference formula becomes

$$y_p \approx y_0 + C(p,1)\Delta y_0 + C(p,2)\Delta^2 y_0 + \dots$$

Newton's backward difference formula may be written

$$y_p \approx y_0 + C(p,1)\nabla y_0 + C(p+1,2)\nabla^2 y_0 + C(p+2,3)\nabla^3 y_0 + \dots$$

We now construct the so called lozenge diagram as shown in Fig. 5.1.

To get an interpolation formula from the lozenge diagram the following rules are observed:

- 1a: For a left to right step, add.
- 1b: For a right to left step, subtract.
- 2a: If the slope of the step is positive, use the product of the difference multiplied with the factor immediately below.

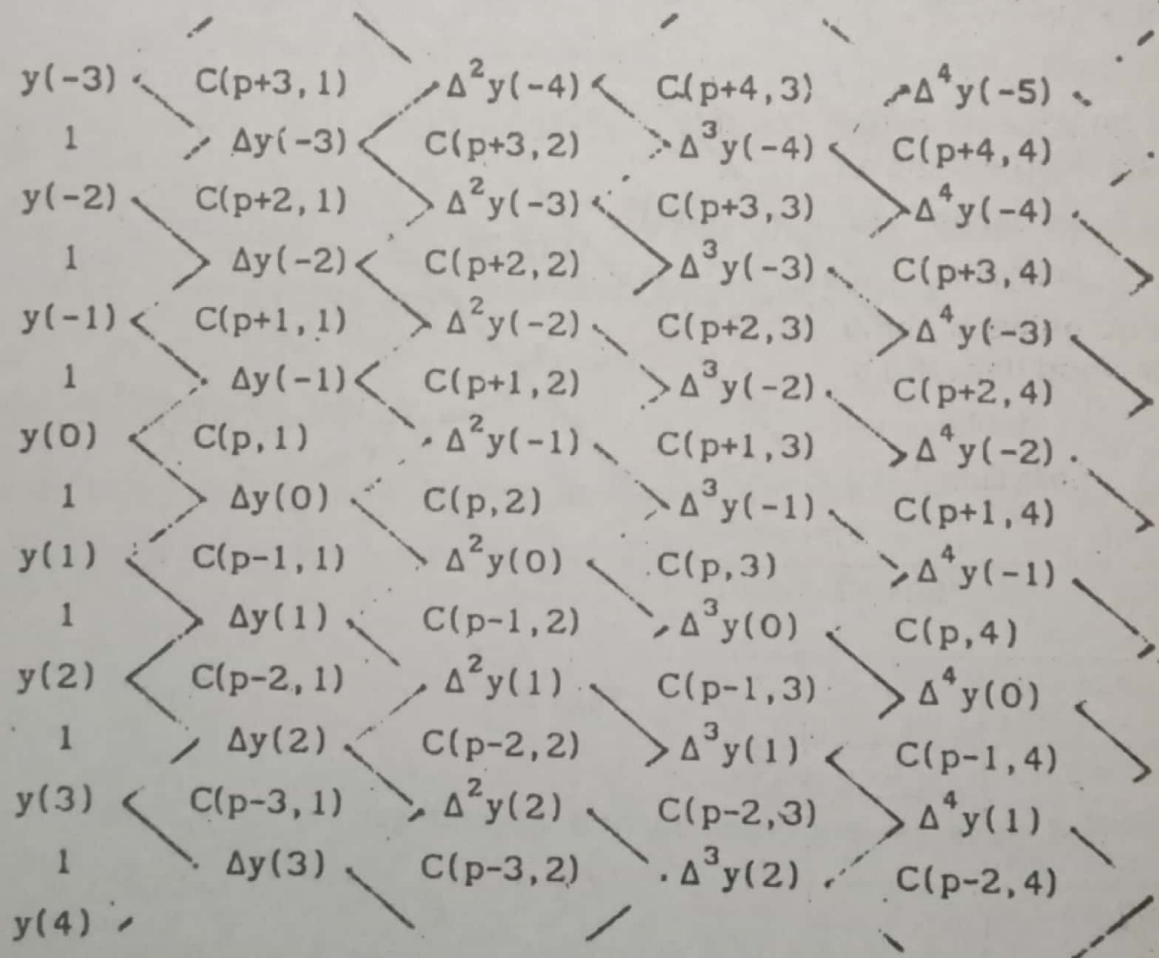


Fig. 5.1 The Lozenge diagram

- 2b: If the slope of the step is negative, use the product of the difference multiplied with the factor immediately above.
- 3a: If the step is horizontal and passes through a difference, use the product of the difference times the average of the factors above and below.
- 3b: If the step is horizontal and passes through a factor, use the product of the factor times the average of the differences above and below.

The validity of the lozenge diagram rests on the facts that (a) the contribution around any closed path is zero, therefore one may deform a path into any other path and (b) if two formulae end at the same place, they are the same, no matter where they started from. The proofs of the above statements may be found in Hamming [6].

Now we make use of the above rules to find various formulae.

1. We start at y_0 and go down and to the right.

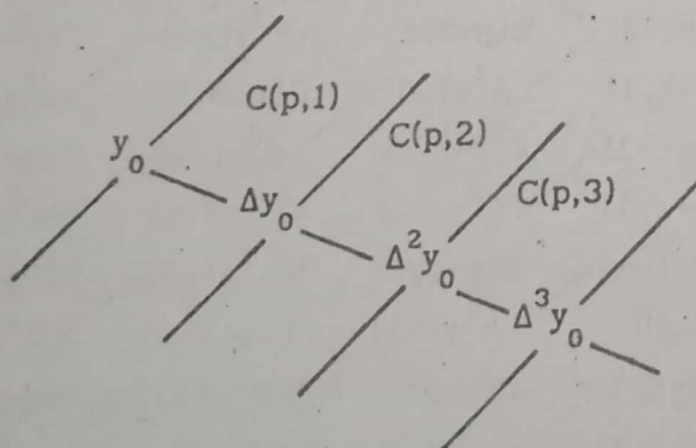


Fig. 5.2

We have to use rules 1a and 2b and get

$$y_p \approx y_0 + C(p,1)\Delta y_0 + C(p,2)\Delta^2 y_0 + C(p,3)\Delta^3 y_0 + \dots$$

which is Newton's forward difference formula.

2. We start at y_0 and move up and to the right

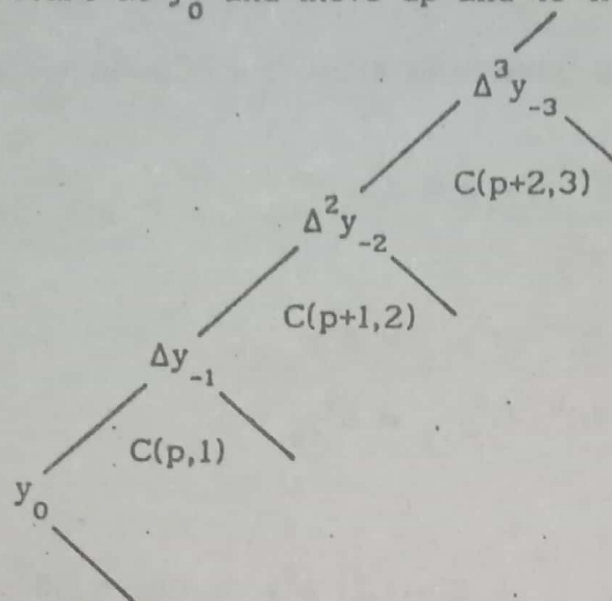


Fig. 5.3

We make use of rules 1a and 2a and get

$$y_p \approx y_0 + C(p,1)\Delta y_{-1} + C(p+1,2)\Delta^2 y_{-2} + C(p+2,3)\Delta^3 y_{-3} + \dots$$

Since $\Delta y_{-1} = \nabla y_0$, $\Delta^2 y_{-2} = \nabla^2 y_0$ etc., we see that the above formula is identical with Newton's backward difference formula.

5.4 GAUSS FORMULAE

If we start at y_0 and zigzag as shown in the Fig. 5.4 we get the following formulae [rules 1a, 2a, 2b]

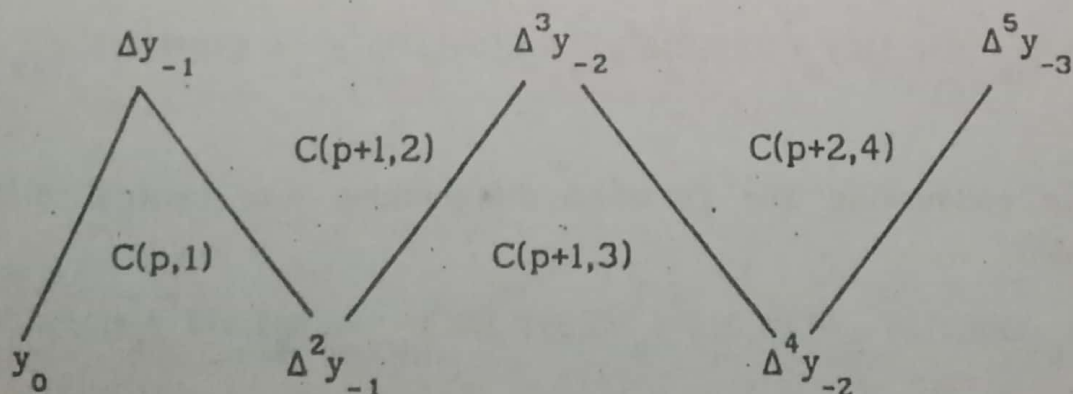


Fig. 5.4

$$y_p \approx y_0 + C(p,1)\Delta y_{-1} + C(p+1,2)\Delta^2 y_{-1} + C(p+1,3)\Delta^3 y_{-2} + C(p+2,4)\Delta^4 y_{-2} \dots$$

Using $\Delta = E^{1/2}\delta$, we can write the above formula in terms of central differences:

$$\Delta y_{-1} = E^{1/2} \delta y_{-1} = \delta y_{-1/2}$$

$$\Delta^2 y_{-1} = \delta^2 y_0$$

$$\Delta^3 y_{-2} = (E^{1/2}\delta)^3 y_{-2} = \delta^3 y_{-1/2}$$

$$\Delta^4 y_{-2} = (E^{1/2}\delta)^4 y_{-2} = \delta^4 y_0 \text{ etc.}$$

Thus we have

$$y_p \approx y_0 + C(p,1) \delta y_{-1/2} + C(p+1,2) \delta^2 y_0 + C(p+1,3) \delta^3 y_{-1/2} + C(p+2,4) \delta^4 y_0 + \dots \quad (5.3)$$

This is known as Gauss backward formula. If we zigzag as shown in Fig. 5.5 and use rules 1a, 2a and 2b,

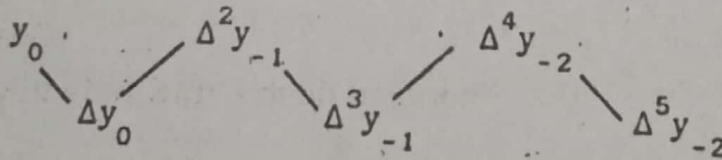


Fig. 5.5

we get the following formula

$$y_p \approx y_0 + C(p,1)\Delta y_0 + C(p,2)\Delta^2 y_{-1} + C(p+1,3)\Delta^3 y_{-1} + C(p+1,4)\Delta^4 y_{-2} + \dots$$

Again converting the forward differences into central differences we get

$$y_p \approx y_0 + C(p,1)\delta y_{1/2} + C(p,2)\delta^2 y_0 + C(p+1,3)\delta^3 y_{1/2} + C(p+1,4)\delta^4 y_0 + \dots \quad (5.4)$$

This is known as Gauss forward formula.

5.5 STIRLING'S INTERPOLATION FORMULA

If we take average of the two Gauss formulae we get

$$y_p \approx y_0 + C(p,1) \frac{\delta y_{-1/2} + \delta y_{1/2}}{2} + \frac{C(p+1,2) + C(p,2)}{2} \delta^2 y_0 \\ + C(p+1,3) \frac{\delta^3 y_{-1/2} + \delta^3 y_{1/2}}{2} + \dots$$

or

$$y_p \approx y_0 + p\mu\delta y_0 + \frac{p^2}{2!} \delta^2 y_0 + \frac{p(p^2-1)}{3!} \mu\delta^3 y_0 + \frac{p^2(p^2-1)}{4!} \delta^4 y_0 + \dots \quad (5.5)$$

This is known as Stirling's interpolation formula. Note that this formula can also be obtained from the lozenge diagram by applying rules 3a and 3b [start at y_0 and go horizontally to the right].

5.6 BESSEL'S INTERPOLATION FORMULA

If we start at the middle of y_0 and y_1 and go horizontally to the right, we get Bessel's formula:

$$y_p \approx \frac{y_0 + y_1}{2} + \frac{C(p,1) + C(p-1,1)}{2} \Delta y_0 + C(p,2) \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \dots$$

$$\text{or } y_p \approx \frac{y_0 + y_1}{2} + (p-1/2)\Delta y_0 + \frac{p(p-1)}{2} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \dots \quad (5.6)$$

The above formula may be written in an equivalent form, using μ, δ notation.

$$y_p \approx \mu y_{1/2} + (p-1/2)\delta y_{1/2} + \frac{p(p-1)}{2} \mu\delta^2 y_{1/2} + \dots \quad (5.6')$$

5.7 EVERETT'S INTERPOLATION FORMULA

In the Gauss's formula

$$y_p \approx y_0 + C(p,1)\Delta y_0 + C(p,2)\Delta^2 y_{-1} + C(p+1,3)\Delta^3 y_{-1} + C(p+1,4)\Delta^4 y_{-2} + \dots,$$

we replace the odd order differences by even order differences as follows:

$$\Delta y_0 = y_1 - y_0$$

$$\Delta^3 y_{-1} = \Delta^2(\Delta y_{-1}) = \Delta^2(y_0 - y_{-1})$$

etc. We now have

$$y_p \approx [1 - C(p,1)]y_0 + [C(p,2) - C(p+1,3)]\Delta^2 y_{-1} + C(p+1,4)\Delta^4 y_{-2} + \dots \\ + C(p,1)y_1 + C(p+1,3)\Delta^2 y_0 + \dots$$

Using $q = 1 - p$, the above result may be written as

$$y_p \approx qy_0 + C(q+1,3)\delta^2 y_0 + C(q+2,5)\delta^4 y_0 + \dots \\ + py_1 + C(p+1,3)\delta^2 y_1 + C(p+2,5)\delta^4 y_1 + \dots \quad (5.7)$$

This is the important Everett's formula. This is useful because it only uses even order central differences and these are sometimes tabulated along with the values of the function.

The two Newton's formulae are useful for interpolation near an end point. Stirling's formula is used for small values of p , say $-1/4 \leq p \leq 1/4$ and Bessel's formula is suitable when p is close to $1/2$, for example $1/4 \leq p \leq 3/4$. Everett's formula is also extensively used.

EXAMPLE 5.3: In the following table values of $y = x + \sin x^2$ are tabulated.

x	y
1.0	1.84147
1.1	2.03562
1.2	2.19146
1.3	2.29290
1.4	2.32521
1.5	2.27807
1.6	2.149355

Construct a difference table and estimate $f(1.04)$, $f(1.57)$, $f(1.28)$, $f(1.324)$.

SOLUTION: The difference table for the above data will be as follows:

y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1.84147					
	0.19415				
2.03562		-0.03831			
	0.15584		-0.01609		
2.19146		-0.05440		0.00136	
	0.10144		-0.01473		0.00305
2.29290		-0.06913		0.00441	
	0.03231		-0.01032		0.003785
2.32521		-0.07945		0.008195	
	-0.04714		-0.002125		
2.27807		-0.081575			
	-0.128715				
2.149355					

(1) To estimate $f(1.04)$, we use Newton's forward formula.

Here $y_0 = 1.84147$

$$p = \frac{x-x_0}{h} = \frac{1.04-1.00}{0.1} = \frac{0.04}{0.1} = 0.4.$$

Hence

$$\begin{aligned} y_p &= y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots \\ &= 1.84147 + (0.4)(0.19415) + \frac{0.4(0.4-1)}{2}(-0.03831) \\ &\quad + \frac{(0.4)(0.4-1)(0.4-2)}{6}(-0.01609) \\ &= 1.922697. \end{aligned}$$

The exact value is 1.922711.

(2) To estimate $f(1.57)$, we use Newton's backward difference formula.

Here $y_0 = 2.149355$

$$p = \frac{x-x_0}{h} = \frac{1.57-1.60}{0.1} = \frac{-0.03}{0.1} = -0.3$$

$$\begin{aligned} y_p &\approx y_0 + p\nabla y_0 + \frac{p(p+1)}{2!}\nabla^2 y_0 + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_0 + \dots \\ &= 2.149355 + (-0.3)(-0.128715) + \frac{(-0.3)(-0.3+1)}{2}(-0.081575) \\ &\quad + \frac{(-0.3)(-0.3+1)(-0.3+2)}{6}(-0.002125) \\ &= 2.19666. \end{aligned}$$

The exact value is 2.196218.

(3) To estimate $f(1.28)$ we first use Stirling's formula, then compare this with the corresponding result of Everett's formula. Here take $x_0 = 1.30$, therefore $p = -0.2$.

Stirling's formula:

$$y_p \approx y_0 + C(p,1) \frac{\Delta y_0 + \Delta y_{-1}}{2} + \frac{C(p+1,2) + C(p,2)}{2} \Delta^2 y_{-1} + \dots$$

Putting $p = -0.2$ and substituting for $y_0, \Delta y_0$ etc., we get

$$y_p \approx 2.29290 + (-0.2) \left(\frac{0.03231 + 0.10144}{2} \right) + \frac{(0.8)(0.8-1)/2 + (-0.2)(-1.02)/2}{2} (-0.06913) \\ = 2.27876457.$$

The exact value is 2.277716.

Now use Everett's formula. We take

$$x_0 = 1.20, x_1 = 1.30, \text{ Therefore } p = 0.8 \text{ and } q = 1 - p = 0.2.$$

$$\text{Now } y_p \approx qy_0 + C(q+1,3)\delta^2 y_0 + \dots + py_1 + C(p+1,3)\delta^2 y_1 + \dots \\ = (0.2)(2.19146) + \frac{(1.2)(1.2-1)(1.2-2)}{6} (-0.05440) \\ + (0.8)(2.29290) + \frac{(1.8)(1.8-1)(1.8-2)}{6} (-0.06913) \\ = 0.4400328 + 1.83763824 = 2.27767104.$$

We note that the result, in this case, is better than the earlier one calculated by the use of Stirling's formula.

EXERCISE 5

1. A function $y = f(x)$ is given by the following table.

x	0	1	2	3	4	5	6
f(x)	2	3	10	29	66	127	218

- By constructing a difference table, determine the order of the most appropriate polynomial approximation to this data. Hence find

- $f(1.5)$ Using Newton's forward formula
- $f(5.5)$ Using Newton's backward formula
- $f(3.5)$ Using i) Stirling's ii) Bessel's
iii) Everett's formulae.

2. From the following data:

x	0	0.2	0.4	0.6	0.8	1.0	1.0
y	-1.651	0.300	2.178	3.505	3.800	2.582	-0.627

a) approximate $y(0.1)$, $y(0.45)$, $y(1.1)$

b) estimate $y(0.25)$ using Newton's forward formula and Stirling's formula.

3. Construct a difference table for the following data, and hence estimate $y(0.5437)$.

x	0.51	0.52	0.53	0.54	0.55
y	0.5292437	0.5378987	0.5464641	0.5549392	0.5633233
x	0.56	0.57			
y	0.5716157	0.5798158			

4. Find the sum of series

$$S_n = 1^2 + 2^2 + \dots + n^2$$

by constructing a difference table and then using Newton's forward formula.

5. Find a formula for the function $f(x)$ given by the following table:

x	0	1	2	3	4	5
f(x)	5.2	8.0	10.4	12.4	14.0	15.2

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x	0	0.2	0.4	0.6	0.8	1.0	1.0
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